For an operational definition of "Pre-Basque": status, scope, utility and reproducible protocols

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Abstract

This text stabilizes the term *Pre-Basque* (Pre-B) as an intermediate analytical category distinct from the *proto-Basque* reconstruction and from attested *modern Basque*. It sets out explicit attribution criteria, a locked operational grid (data formats, absolute filters, scoring, statistical tests) and traceability protocols designed to make any Pre-B assignment testable and reproducible. Minimal micro-cases illustrate practical application. The aim is to provide a transparent, testable, and interdisciplinary methodological instrument for handling residual ancient linguistic material in the Basque area.

Keywords: Pre-Basque; proto-Basque; linguistic stratigraphy; toponymy; reproducibility; protocol

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1. Introduction

The vocabulary used to discuss ancient layers in the Basque domain suffers from terminological drift. “Pre-Basque” is variously used as a chronological label, as a synonym for *proto-Basque*, or as a convenient tag for apparently archaic forms. This indeterminacy undermines methodological clarity and comparability of results. This text proposes an operational definition of Pre-B, explicit delimitation criteria, and a complete reproducible analysis protocol to be published together with datasets and scripts. The goal is methodological: to provide a descriptive, falsifiable tool rather than a definitive reconstruction.

2. Problem statement

Three recurring problems motivate the introduction of a Pre-B level:

- **Dispersion of problematic forms**: certain toponyms and morphological residues do not fit standard proto-Basque reconstructions and are often treated as exceptions.
- **Morphological opacity**: residual prefixes, suffixes and composition patterns in modern Basque suggest a partially erased earlier system.
- **Terminological and methodological drift**: conflating reconstructive hypotheses with intermediate descriptive levels leads to ad hoc protocols and non-reproducible attributions.

These issues call for a finer analytical stratigraphy: Pre-B is proposed as an explanatory layer that organizes residual formal and toponymic material into partial, heuristically productive systems.

3. Operational definition of Pre-Basque

Pre-Basque is defined as an **intermediate analytical level** made up of a bundle of traits (segments, morphs, toponymic schemata) that simultaneously satisfy three conditions:

1. **Not directly observable** in modern varieties.
2. **Inadequate for usual proto-Basque reconstructions**.
3. **Analytical coherence**: traits cluster into regular patterns enabling explicit rules and testable predictions.

Pre-B is not a reconstructible language state but a *heuristic tool* subject to empirical falsification.

4. Conceptual distinction: three analytical planes

Modern Basque

Attested varieties documented by contemporary linguistics.

Proto-Basque

A comparative reconstruction aiming to restore a hypothetical earlier state.

Pre-Basque (Pre-B)

An intermediate analytical level accounting for residual forms and toponymic patterns escaping classical reconstruction logic.

This tripartition is analytical, not strictly chronological.

5. Delimitation criteria

Robust Pre-B attribution relies on explicit, reproducible criteria.

Formal criteria

- Recurrent correspondences not explained by reconstructed laws.
- Residual morphological motifs forming patterns.
- Repeated combinatorial distributions.

Toponymic and onomastic criteria

- Coherent toponymic sets.
- Anthroponymic/hydronymic forms pointing to non-transparent schemata.
- Geo-historical concordance.

Internal coherence and testability

- Capacity to form a partial system.
- Predictive power.
- Falsifiability.

6. Reproducible protocols: data formats, grid and governance

Canonical input format

Required CSV header: `form|source|region|reliability`.
Optional: `lat|lon|date_attestation|variant|notes`.

Operational checklist

1. Normalization.
2. Exploratory skeleton extraction.
3. Guided vocalizations.
4. Mandatory triple filter.
5. Apply FNA (absolute negative filters).

6. Scoring and verdict.
7. Statistical tests.
8. Archiving and traceability.

Absolute Negative Filters

Immediate exclusion criteria:

- Impossible initials: `m-, p-, f-, j-, tx-, tt-, ll-, ñ-, x-, r-, sC-`.
- Impossible segments: `f, j, tx, tz, tt, ts, ll, ñ, x, h_morphological`.
- Impossible structures: Latin/Romance clusters; non-vocalized final occlusives; non-derivable rising diphthongs.
- Absolute lexical exclusions.

Weighting system

Weights: phonology 2; morphology 2; distribution 2; skeleton 1; external 1; borrowing -2.
Thresholds:

- A: score ≥ 6.
- B: score $= 2$.
- C: score $= 1$.

Statistical tests

Randomization of vocalizations (seed = 42). Interpretation: $p < 0.05 \rightarrow$ unlikely coincidence.

Output format

```
Form | Source / region / reliability | Reconstructions (<=2) + rule |
Morphology | Borrowing | Distribution | Coincidence test |
Score & verdict | Justification | Recommended action | Trace
```

Governance

Version the configuration file; document modifications; log each run.

7. Operational illustrations: micro-cases

Micro-case 1 — *ibar*

Score = 6 (A). Simulation: $p \approx 0.004$.

Micro-case 2 — morphological alternation

Exploratory skeleton + two vocalizations suggest a formerly productive suffix.

8. Empirical validation methodology

1. Systematic inventory.
2. Classification by criteria.
3. Robustness tests.
4. Interdisciplinary confrontation.
5. Versioning and transparency.

9. Theoretical and practical implications

- **Historical reconstruction**.
- **Toponymy and onomastics**.
- **Comparative methodology**.
- **Scientific communication**.

10. Limits, risks and safeguards

- Distinguish hypothesis from proof.
- Avoid anachronistic narratives.
- Make criteria explicit.
- Keep the category contestable.
- Respect cultural sensitivity.

11. Manual plan and annexes

The manual will include theory, formats, inventories, tables, case studies, tests and scripts. It will rely on a corpus of approximately 300 attested forms, processed through our standardized protocol.

12. Conclusion

Introducing *Pre-Basque* as an operational category addresses a concrete methodological need. Its usefulness will be measured by its capacity to yield simpler, more predictive and reproducible analyses.

Selected annotated bibliography

- Trask (1997). *The History of Basque.*
- Hualde (2003). *Basque Phonology and Morphology.*
- Lakarra (2004).
- Michelena (1961). *Fonética histórica vasca.*
- Renfrew & Bahn (1987).
- Campbell (2004).

Annex A — Example analysis sheets

Each sheet follows the grid fields.

1. *ibar*

Form (attested):	*ibar*
	Source / region / reliability:
toponymy; modern Basque dictionary; Pays Basque; **high**.	*ibar*; *ibara.*
	Skeleton extraction:
Reconstructions (<=2):	

B-R.	(1) *ibar*; (2) *ibara*.
Plausible vocalizations:	Morphology:
nominal root “valley”;	none.
determinative **-a**.	Distribution:
Borrowing indicators:	
widespread; stable	p ≈ 0.004.
meaning “valley”.	Score & verdict:
Coincidence test:	
+7 → **A**.	phonology + morphology + distribution coherent.
Short justification:	Recommended action:
classify as A; attach	rules invoked; comparanda; simulation; weighting.
attestations.	
Trace:	

2. *mendi*

Form (attested):	*mendi*
Source / region / reliability:	dictionary; toponymy; **high**.
Reconstructions:	*mendi*; *mendia*.
Skeleton extraction:	m-n-d.
Plausible vocalizations:	(1) *mendi*; (2) *mendia*.
Morphology:	productive root; **-a**.
Borrowing indicators:	none.
Distribution:	widespread; stable meaning.
Coincidence test:	not required.
Score & verdict:	**+8** → **A**.
Short justification:	lexicalized; productive; broad attestation.
Recommended action:	classify as A.
Trace:	justified vocalizations; weighting.

3. *otso*

Form (attested):	*otso*
Source / region / reliability:	dictionary; lexicon; toponymy; **high**.
Reconstructions:	*otso*; *otsoa*.
Skeleton extraction:	-t-s.
Plausible vocalizations:	(1) *otso*; (2) *otsoa*.
Morphology:	root “wolf”; **-a**.
Borrowing indicators:	none; Aquitanian comparanda.

Distribution:	ancient + modern attestations.
Coincidence test:	qualitative.
Score & verdict:	$\mathbf{+8 \rightarrow A}$.
Short justification:	lexicalized; historical continuity.
Recommended action:	classify as A.
Trace:	comparanda; rules; weighting.

Each sheet records the decision chain and recommended next steps.